

Lightweight Anomaly Detection for Campus Networks

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Abstract

Campus networks carry a mix of student, research and administrative traffic, which makes unusual behaviour hard to spot. We present a lightweight detector that learns what normal traffic looks like for each building and flags connections that fall far outside it. On two weeks of anonymised flow records, the detector found 92% of injected test incidents with a false alarm rate below 1%, while running on a single low-cost server.

1 Introduction

Most intrusion detection systems are built for company networks with predictable traffic. University campuses are different: thousands of personal devices join and leave every day, and research labs regularly produce traffic spikes that look like attacks. In this paper we ask a simple question: can a small, explainable model keep up with this variety without a dedicated security team to tune it?

Key idea: model each building's normal traffic separately, then compare.

Dataset	Detection rate	False alarms
Week 1 (lab buildings)	91%	0.8%
Week 2 (hostels)	93%	0.7%

Code: <https://example.org/campus-detector>

1 Flow records were anonymised before analysis; no packet contents were stored.